

Between Multiple Choice and Open Response: Evaluating LLMs with Option-Free Deterministic Scoring

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🇲🇾 Project Page: <https://github.io>

Abstract

Benchmark scores are often treated as evidence of LLM competence, but the test itself can help produce the score. MCQs provide objective and scalable scoring, but their answer options can turn recall into recognition through answer-option scaffolding. Open-ended constructed response questions (CRQs) remove answer options, but their scores often depend on human rubrics or LLM judges, introducing judge-dependent scoring. This creates a trade-off between option-free elicitation and objective scoring. We study whether this trade-off can be addressed through **Constrained Structured Response Questions (CSRQs)**, a structured-response protocol that removes final answer options while preserving deterministic scoring. CSRQs use canonical multi-select, ordering, and matching outputs, not as new item archetypes, but as auditable answer spaces for exact verification. We evaluate CSRQs using converted knowledge benchmarks and *MyCulture*, to our knowledge the first Malaysia-centered cultural benchmark in Bahasa Malaysia, with parallel MCQ, CSRQ, and CRQ formats. Across benchmarks and model families, measured scores change when the evaluation instrument changes, *i.e.*, scores drop after answer options are removed, and CRQ scores vary across judges. These findings suggest that benchmark scores should be interpreted as products of both model capability and the test used to measure it.

1 Introduction

Benchmark scores are often treated as evidence of model competence, yet they are also shaped by the instrument used to elicit and score answers. Multiple choice questions (MCQs) are scalable and deterministically scored, but visible answer candidates can support recognition, elimination, and chance success (Burton, 2001;

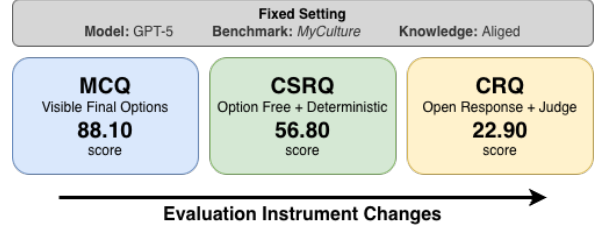


Figure 1: **Does a high LLM benchmark score reflect model capability alone, or also the evaluation instrument?** MCQs are easy to score but expose answer choices; open-ended CRQs remove them but introduce judge dependence. We therefore propose CSRQ, which removes candidate final-answer combinations while retaining deterministic scoring. On *MyCulture*, our Malaysia-centered Bahasa Malaysia benchmark, the same GPT-5 and underlying knowledge yield markedly different scores.

Balepur et al., 2025; Raman et al., 2025; Chandak et al., 2025). Open ended constructed response questions (CRQs) remove these candidates, but their scores can depend on human or LLM judges, including the judge model and rubric interpretation (Livingston, 2009; Zheng et al., 2023; Bavaresco et al., 2025; Shi et al., 2025; Watts et al., 2024). Thus, a benchmark score may reflect both model capability and the evaluation instrument.

We study whether eliciting answers without final options can be combined with deterministic scoring through **Constrained Structured Response Questions (CSRQs)**. Rather than selecting a final answer from composite candidates, a model constructs a canonical structured response in the form of a subset, ordering, or matching that can be verified exactly. For example, an MCQ may ask a model to choose between “A. Statements 1 and 2” and “B. Statements 1 and 3.” The corresponding CSRQ presents the individual statements but requires the model to output the correct subset, “1 and 2,” without exposing candidate answer combinations.

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To examine this under controlled conditions, we introduce *MyCulture*, a new Malaysia-centered cultural benchmark in Bahasa Malaysia created in this work. *MyCulture* is designed specifically as a parallel-format testbed: each underlying cultural knowledge item is instantiated in MCQ, CSRQ, and CRQ formats. This design holds the tested knowledge constant while varying the evaluation instrument, enabling direct analysis of sensitivity to visible answer options and judge-based scoring, as illustrated in Figure 1. CSRQ is not proposed as a new human test item archetype; multi selection, ordering, and matching have long existed in assessment. As illustrated in Appendix Figure 7, CSRQs occupy a middle ground between conventional MCQs and open-ended CRQs: their final responses are constructed rather than selected from enumerated final-answer combinations, while their constrained answer spaces remain deterministically scorable. Our contribution is to use these structured output spaces as canonical answer spaces for LLM evaluation.

CSRQ is not intended to prove genuine understanding, measure higher order comprehension, or replace existing benchmarks. Instead, it isolates how measured scores change when final answer options are removed while deterministic scoring is retained. The difference between MCQ and CSRQ scores therefore measures sensitivity to removing answer options, whereas variation across CRQ judges measures sensitivity to the judge used for scoring. Together, these analyses test the broader claim that benchmark scores are outcomes of both the model and the instrument, not properties of the model alone.

We evaluate this claim in two settings. First, we convert items from existing knowledge benchmarks into CSRQ format to test whether sensitivity to answer options extends beyond our cultural testbed. Second, we evaluate *MyCulture* across parallel MCQ, CSRQ, and CRQ formats to study how evaluation instruments shape measured cultural competence.

Our contributions are as follows:

- (i) We frame knowledge oriented LLM evaluation, with cultural competence as a central testbed, as an instrument validity problem and identify two evaluation induced illusions: one from answer options in MCQs and one from judges in CRQs.
- (ii) We study whether the useful functions of

MCQs and CRQs can be recomposed through **Constrained Structured Response Questions** (CSRQs), an auditable protocol that elicits answers without final options and uses deterministic scoring, and provide a structured workflow for converting suitable benchmark items into CSRQs.

- (iii) We instantiate and evaluate CSRQs through converted knowledge benchmarks and *MyCulture*, a cultural testbed centered on Malaysia with parallel MCQ, CSRQ, and CRQ formats, showing that measured competence changes when evaluation instrument changes.

2 Constrained Structured Response Questions (CSRQs)

2.1 Motivation

We formalize CSRQ around the evaluation-format tradeoff introduced above. MCQs provide objective scoring, but expose final answer options. CRQs remove final answer options, but require judge-dependent scoring. CSRQ studies whether these useful functions can be recomposed into an intermediate evaluation instrument.

Answer-option scaffoldin. MCQs combine the question with visible answer candidates. This design is useful for objective scoring, but it can also turn recall into recognition. A model may use answer options as hints, rely on elimination, or succeed through partial knowledge rather than independently producing the target answer. Existing MCQ benchmarks also have non-trivial random baselines, often around 25% for four-option settings and higher for some structured formats (Li et al., 2023; Hendrycks et al., 2020; Poh et al., 2024). We refer to this confound as *answer-option scaffoldin*, where the measured score partly reflects answer-option scaffolding rather than model competence alone.

Judge-dependent scoring. Open-ended constructed response questions (CRQs) remove final answer options and therefore support option-free elicitation. *Final-answer-option-free* (denoted as *option-free* hereafter) elicitation means that no candidate final answers are explicitly enumerated. CSRQs provide item-specific statements or descriptions as response primitives, but do not expose candidate subsets, orderings, or matchings

from which the model selects. However, their scores depend on how responses are judged. Human grading is costly and difficult to scale, while LLM judges can vary with prompt wording, rubric interpretation, judge model, and model capability (Shi et al., 2025; Bavaresco et al., 2025). We refer to this confound as *judge-dependent scoring*, where the measured score reflects not only the evaluated response but also the scoring instrument.

These two confounds suggest that the key question is not whether MCQs or CRQs are better. The key question is whether their useful functions can be recomposed. *Can evaluation remove final answer options like CRQs, while retaining deterministic scoring like MCQs?*

2.2 Formulation

We study this recomposition through **Constrained Structured Response Questions (CSRQs)**, an auditable structured-response protocol that combines option-free elicitation with deterministic scoring. CSRQ is not proposed as a new human-test item archetype. Multi-selection, ordering, and matching have long existed in educational and psychological assessment. Our contribution is to use these structured output spaces as canonical answer spaces for LLM evaluation.

Formally, a CSRQ is a tuple (Q, S, A) , where Q is the question prompt, $S = \{s_1, \dots, s_M\}$ is a set of M provided statements or descriptions, and A is the ground-truth answer expressed in a canonical structure. Unlike an MCQ with a visible option set $C = \{c_1, \dots, c_K\}$, a CSRQ does not expose final answer candidates. The model must construct the answer $\hat{A}$ directly from the task specification and the provided descriptions.

We instantiate CSRQs through three structured response variants.

- (i) **Multi Select (MS)** requires the model to output all statements in S that satisfy the condition specified in Q . Given $S = \{s_1, s_2, s_3, s_4\}$, a valid prediction may be $\hat{A} = \{s_1, s_3\}$.
- (ii) **Ordering (O)** requires the model to output an ordered sequence over S according to the relation specified in Q , such as chronology, procedure, or causal progression. Given $S = \{s_1, s_2, s_3, s_4\}$, a valid prediction may be $\hat{A} = (s_2, s_4, s_1, s_3)$.
- (iii) **Matching (M)** requires the model to

construct pairwise associations between two statement sets. Given $S_A = \{s_{a_1}, s_{a_2}, s_{a_3}\}$ and $S_B = \{s_{b_1}, s_{b_2}, s_{b_3}\}$, a valid prediction may be $\hat{A} = \{(s_{a_3}, s_{b_2}), (s_{a_1}, s_{b_3}), (s_{a_2}, s_{b_1})\}$.

2.3 Functional Properties

CSRQs are designed around two functional properties, which we describe as follows:

Option-free elicitation. CSRQs do not provide final answer candidates. The statements in S are descriptive units used to define the task, while MCQ options are candidate outputs that may directly contain the correct answer. In an MCQ, the model selects from visible answer candidates. In a CSRQ, the model constructs a structured answer such as a subset, an ordering, or a matching.

Deterministic scoring. CSRQs preserve the main scoring function of MCQs without relying on visible answer options. Since the admissible output is a structured object, a model response can be canonicalized and compared against a predefined key. Let $\hat{A}$ denote the model prediction and A denote the ground-truth answer. We define a canonicalization function $f(\cdot)$ that normalizes equivalent surface forms. Strict scoring is defined as

$$\text{Score}(\hat{A}) = \mathbb{I} [f(\hat{A}) = f(A)],$$

where $\mathbb{I}[\cdot]$ returns 1 if the canonicalized prediction exactly matches the canonicalized answer, and 0 otherwise. The score is determined by the response structure, not by human preference, rubric interpretation, or judge-model sensitivity.

2.4 Validity Considerations

From a psychometric perspective, evaluation instruments shape the constructs and response processes being measured (Messick, 1995; Kane, 2013). CSRQs strengthen construct validity by reducing construct-irrelevant variance associated with option-order effects, distractor quality, and rater-dependent partial-credit assignment (Rodríguez, 2003; Jonsson and Svingby, 2007). Their explicitly constrained answer space also improves content control by allowing the assessed propositions and difficulty level to be systematically specified. Compared with conventional CRQs, CSRQs therefore provide a more standardized and diagnostically interpretable measure of model knowledge and constraint satisfaction. Nevertheless,

content coverage and robustness to superficial format variations should still be examined to ensure that the constrained response space adequately represents the intended capability (Kane, 2013; Sireci and Faulkner-Bond, 2014).

2.5 Chance-Success Analysis

We characterize random-success probabilities under exact structural scoring. This analysis is deliberately narrow. It does not measure semantic difficulty, cognitive complexity, or genuine understanding. It only describes how likely a random answer is to exactly match the canonical answer under each format.

For a K -choice MCQ with one correct answer, random guessing succeeds with probability

$$P_{\text{MCQ}}(\hat{A} = A \mid Q, C) = \frac{1}{K}.$$

For a Multi Select CSRQ over M statements, the admissible answer space contains all non-empty subsets of S . The random-success probability is

$$P_{\text{MS}}(\hat{A} = A \mid Q, S) = \frac{1}{2^M - 1}.$$

For an Ordering CSRQ, the admissible answer space contains all permutations of S . The random-success probability is

$$P_{\text{O}}(\hat{A} = A \mid Q, S) = \frac{1}{M!}.$$

For a Matching CSRQ with two equal-sized sets S_A and S_B , the admissible answer space contains all bijections from S_A to S_B . The random-success probability is

$$P_{\text{M}}(\hat{A} = A \mid Q, S_A, S_B) = \frac{1}{M!}.$$

These baselines quantify one source of option-removal sensitivity under exact scoring. They should not be interpreted as evidence that CSRQs measure deeper understanding or competency.

3 Empirical Validation

We evaluate whether CSRQ can serve as a feasible intermediate instrument between MCQ and CRQ under a closed-book, tool-free setting, without web search, retrieval, or external tool use. The goal is not to prove that CSRQ measures genuine understanding, nor to claim that CSRQ replaces

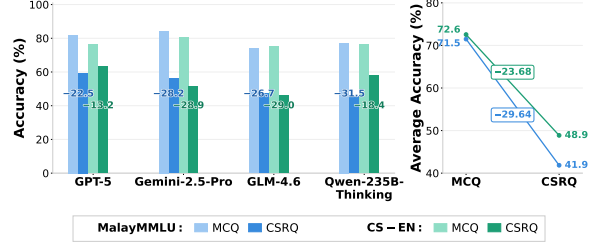


Figure 2: **Option-removal sensitivity in existing benchmarks.** Left: MCQ and CSRQ scores for representative models on MalayMMLU and CS-EN. Right: average MCQ and CSRQ scores across all evaluated models. The consistent drop after removing final answer options shows option-removal sensitivity under deterministic scoring, rather than evidence that CSRQ measures deeper understanding.

existing benchmarks. Instead, we test whether measured scores change when final answer options are removed while deterministic scoring is preserved, and whether open-ended CRQ scores vary under judge-dependent grading.

Evaluated Models. We evaluate a diverse set of closed- and open-source LLMs spanning multiple model families, scales, and reasoning modes, including OpenAI (OpenAI, 2025), Gemini (Comanici et al., 2025), GLM (Zeng et al., 2025), and Qwen3 (Yang et al., 2025). Where available, both reasoning and instruction-tuned variants are included. Models range from compact models below 10B parameters to large-scale models above 100B parameters. Full list is provided in Appendix D.

3.1 Instrument Sensitivity in Existing Benchmarks

We first examine whether option-removal sensitivity appears beyond *MyCulture*. We use two complementary strategies. First, we extract existing multi-answer MCQ items and remove final answer options while preserving the underlying question intent. Second, we convert representative MCQ- and CRQ-based benchmarks into CSRQ format to study how measured scores change under option-free elicitation and deterministic scoring.

3.1.1 Extracting Existing Multi-Answer MCQs

We select MalayMMLU (Poh et al., 2024) and CS-Bench (Song et al., 2025) because they contain multi-answer MCQ items that can be directly reformulated as CSRQs. This setting allows a con-

MMLU			
Model	MCQ	CSRQ	Δ
GPT-5	92.50	84.64	7.86
Gemini-2.5-Pro	<u>91.06</u>	<u>78.35</u>	12.71
GLM-4.6	84.38	80.59	3.79
Qwen3-235B-Thinking	90.39	80.90	8.61

HealthBench-Hard			
Model	MCQ	CSRQ	CRQ
GPT-5	96.17	83.33	50.84
Gemini-2.5-Pro	<u>93.99</u>	75.68	31.07
GLM-4.6	95.63	<u>77.87</u>	<u>32.34</u>
Qwen3-235B-Thinking	95.08	<u>78.42</u>	<u>41.18</u>

Table 1: **Supporting diagnostic on converted benchmarks.** For MMLU, Δ is the absolute score difference between MCQ and CSRQ evaluation. For HealthBench-Hard, we construct matched MCQ, CSRQ, and CRQ versions from the same underlying items. CRQ scores are judged using Qwen3-235B-Instruct.

trolled comparison between MCQ and CSRQ evaluation because the underlying knowledge remains fixed while the final answer options are removed.

Insights. Figure 2 compares model performance under MCQ and CSRQ evaluation on the same underlying items. Across models and benchmark subsets, scores consistently decline after final answer options are removed. This pattern appears across model families and parameter scales, including models above 100B parameters.

This drop should not be interpreted as proof that models lack knowledge. It indicates that measured performance changes when the evaluation instrument changes. In our work, this is option-removal sensitivity. MCQs expose final answer candidates, while CSRQs require models to construct the answer from a larger structured answer space. The observed gap therefore provides supporting evidence that MCQ scores may partly benefit from answer-option scaffolding. Full results are reported in Appendix E.

3.1.2 Converting Existing Benchmarks

Direct extraction is only possible when benchmark items already contain a suitable multi-answer structure. To broaden the diagnostic, we convert representative benchmarks into matched evaluation formats. We use MMLU (Hendrycks et al., 2020) and HealthBench-Hard (Arora et al., 2025) as complementary cases. MMLU allows us to examine how scores change after final answer options are removed. HealthBench-Hard is origi-

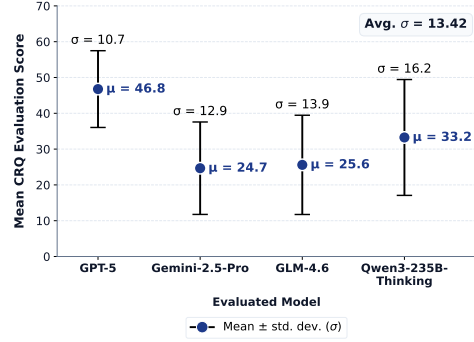


Figure 3: **CRQ evaluation scores under different LLM judges.** Higher standard deviation (σ) indicates greater sensitivity to judge selection. All evaluated models are reasoning models. Full detailed can be found at Appendix J.

nally evaluated using open ended CRQs; we construct matched CSRQ and MCQ versions from the same underlying items. This enables a direct comparison among MCQ, CSRQ, and CRQ evaluation in a non cultural domain while varying answer option exposure and scoring procedure.

Insights. Table 1 provides two complementary diagnostics. On MMLU, scores decrease when MCQ items are converted into CSRQs, consistent with sensitivity to removing final answer options. On HealthBench-Hard, all three evaluated models score highest in the MCQ setting, lower in the CSRQ setting, and lowest in the CRQ setting. This reproduces the three format pattern observed on *MyCulture* in a non cultural benchmark. We do not interpret this ordering as evidence that one format is universally easier, harder, or more valid. Rather, it shows that measured performance changes when answer option exposure and scoring procedures change.

To further examine judge-dependent sensitivity, Table 10 evaluates the same CRQ outputs using different LLM judges under the same rubric. Scores vary substantially across judges, with an average standard deviation of 13.42%. This supports the judge-dependent scoring. The measured score can depend on the scoring instrument, not only on the evaluated response. Full experimental results are reported in Appendix E.

3.2 MyCulture as a Cultural Testbed

We next instantiate the same instrument-validity question in cultural evaluation. Cultural competence is a high-stakes testbed because benchmark scores are often interpreted as evidence that mod-

Model	MCQ	CSRQ		CRQ	
		L	S	L	S
Random	25.53	40.33	4.28	–	–
GPT-5 [†]	88.14	78.10	56.78	68.41	22.90
GPT-4o	90.28	<u>78.00</u>	<u>56.01</u>	68.61	21.14
Gemini-2.5-Pro [†]	85.56	77.78	55.68	66.87	20.26
GLM-4.6 [†]	84.46	76.37	53.38	63.20	17.74
Qwen3-235B-Thinking [†]	61.01	62.62	42.12	50.13	18.51
Qwen3-235B-Instruct	80.12	73.55	51.29	62.93	18.34
SeaLLMs-v3-7B	29.98	52.96	26.30	47.62	7.86
SEA-LION-v4-32B	83.25	76.16	53.54	62.55	18.23

Table 2: **Models evaluated on MyCulture.** L denotes loose scoring, and S denotes strict scoring; scoring details and full results are provided in Appendix F. **Bold** indicates best performance and underline indicates second-best. [†] denotes reasoning models, and by default the reasoning effort is medium where available.

els understand local norms, practices, and values. We do not claim that one benchmark covers all cultural knowledge. Instead, we built *MyCulture* as a controlled Malaysia-centered setting for studying whether measured cultural competence changes across MCQ, CSRQ, and CRQ instruments.

To our knowledge, *MyCulture* is the first Malaysia-centred cultural benchmark in Bahasa Malaysia. It covers six cultural dimensions, including arts, attire, customary practices, entertainment, food, and religions. The benchmark comprises 5,463 questions in total, with parallel MCQ, CSRQ, and CRQ formats constructed to preserve the underlying knowledge while changing the evaluation instrument.

MyCulture is built from reliable Malaysian cultural sources and academic materials. LLMs generate MCQ, CSRQ, and CRQ items, then verify format and source-supported answers; CSRQ answers must be objectively parseable. Strong LLMs flag potentially ambiguous items for manual review, while 5% undergoes auditing by qualified Malaysian residents. The construction of *MyCulture* can be found in Appendix I.

Insights. Table 2 shows that model scores differ substantially across MCQ, CSRQ, and CRQ formats. MCQ scores are generally higher than strict CSRQ scores, showing that measured cultural competence changes when final answer options are removed. CRQ strict scores also differ substantially under judge-based grading, showing a different source of measurement sensitivity. The purpose of this result is not to rank formats by dif-

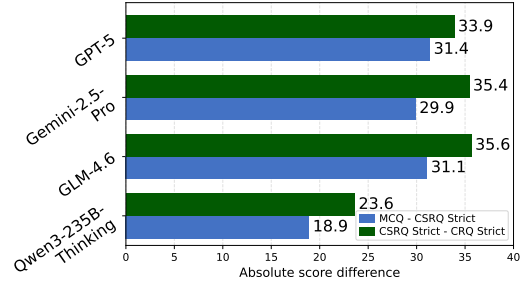


Figure 4: **Instrument gaps on MyCulture across models.** Each bar shows the absolute score change between two evaluation instruments. The MCQ-CSRQ gap reflects option-removal sensitivity, while the CSRQ-CRQ gap reflects the shift from deterministic structured scoring to open-ended judge-based scoring.

Model	Performance Diff. ($\Delta = \text{MCQ} - \text{CSRQ}$)		
	Multi Select	Ordering	Matching
GPT-5	20.43	38.53	<u>37.80</u>
Gemini-2.5-Pro	22.29	<u>33.90</u>	35.38
GLM-4.6	24.86	30.48	39.86
Qwen-235B-Thinking	<u>26.58</u>	17.64	10.24
Average Δ	23.54	30.14	<u>30.82</u>

Table 3: **Performance difference (Δ) between MCQ and CSRQ** across Multi-Select, Ordering, and Matching types on *MyCulture*. **Bold** indicates the largest difference and underline indicates the second-largest.

ficulty, but to show that the evaluation instrument changes the measured score.

To make the instrument effect more explicit, Figure 4 visualizes the absolute score gaps between MCQ, CSRQ-strict, and CRQ-strict evaluation across models. The MCQ-CSRQ gaps are substantial across visualized models, indicating that measured cultural competence decreases when the final answer-option layer is removed while deterministic scoring is preserved. Besides that, we provide a model breakdown of performance degradation in Table 3. The performance degradation is consistent across all CSRQ variants. Full details are available in Table 15 (Appendix G).

Additionally, Table 7 presents a manual error analysis of 120 incorrect responses per model, evenly sampled across the three question formats. On average, most errors are attributable to answer content issues (88.8%), whereas output-format errors account for only 11.3%. The latter mainly arise from response incompleteness, excessively long reasoning traces, or failure to follow the required answer format. Full details are provided in Table 9 (Appendix H).

Table 4: **Example questions across the three evaluation formats.** The three questions assess the same underlying knowledge about the traditional Lansaran game of the Murut community, but differ in how the answer is elicited and scored.

Type	Example
MCQ	Question: Apakah kenyataan yang betul mengenai permainan Lansaran dan budaya suku kaum Murut? Statements: I. Lansaran dibina secara individu tanpa melibatkan masyarakat. II. Permainan Lansaran memerlukan ukuran kawasan antara 8 hingga 30 kaki persegi. III. Kayu “Selangan Batu” digunakan dalam pembinaan lantai Lansaran kerana ketahanannya. IV. Lansaran hanya untuk kanak-kanak dan tidak melibatkan orang dewasa. Options: A. II, III B. I, IV C. I, II, IV D. II, I Answer: A
CSRQ	Question: Apakah kenyataan yang betul mengenai permainan Lansaran dan budaya suku kaum Murut? A. Lansaran dibina secara individu tanpa melibatkan masyarakat. B. Permainan Lansaran memerlukan ukuran kawasan antara 8 hingga 30 kaki persegi. C. Kayu “Selangan Batu” digunakan dalam pembinaan lantai Lansaran kerana ketahanannya. D. Lansaran hanya untuk kanak-kanak dan tidak melibatkan orang dewasa. Answer: B, C
CRQ	Question: Huraikan ciri-ciri fizikal, cara pembinaan dan peranan komuniti dalam permainan tradisional Lansaran suku kaum Murut berdasarkan amalan tradisi yang dinyatakan. Rubric: 1. Tempat permainan Lansaran berukuran antara lapan (8) hingga 30 kaki persegi. 2. Lansaran dapat menampung lebih kurang 30 orang pemain. 3. Lansaran dibina secara bergotong-royong di ruang khas rumah panjang suku kaum Murut. 4. Beberapa batang kayu bulat yang keras dan liat sebesar lengan orang dewasa dikenali sebagai kayu “Selangan Batu”. 5. Kayu “Selangan Batu” diikat dengan rotan saga bagi menampung lantai Lansaran. 6. Lantai Lansaran dibina sekaki ke bawah daripada paras lantai rumah secara tergantung.

Ranking pair	Spearman’s ρ	Kendall’s τ
MCQ–CSRQ–Strict	0.95	0.86
MCQ–CRQ–Strict	0.74	0.50
CSRQ–Strict–CRQ–Strict	0.79	0.64

Table 5: **Rank consistency across MyCulture evaluation instruments.** Correlations are computed over model rankings derived from strict-scoring results.

The CSRQ-CRQ gaps are also large, showing that scores change again when evaluation moves from deterministic structured verification to open-ended judge-based scoring. This pattern supports our central claim that benchmark scores are not only properties of model capability, but also depend on the evaluation instrument used to measure that capability.

Although absolute scores change across instruments, MCQ and CSRQ-Strict produce highly consistent model rankings ($\rho = 0.95$, $\tau = 0.86$; Table 5). CRQ-Strict has lower agreement with both MCQ ($\rho = 0.74$, $\tau = 0.50$) and CSRQ-Strict ($\rho = 0.79$, $\tau = 0.64$), indicating that judge-based CRQ evaluation changes the relative ordering of some models, particularly beyond the top-

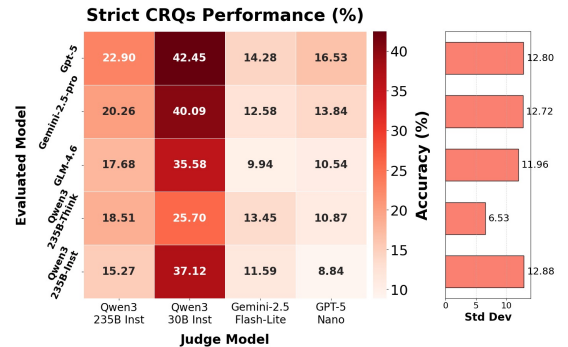


Figure 5: **Judge-induced sensitivity in MyCulture CRQ evaluation.** The same CRQ outputs are evaluated by different LLM judges under the same rubric. The observed variation shows that open-ended cultural evaluation can be sensitive to judge selection.

performing models.

We also evaluate the same CRQ outputs under different LLM judges. Figure 5 shows substantial variation across judges even under the same rubric and evaluation setting. This supports the judge-dependent scoring in the cultural testbed. That is, open-ended cultural scores can reflect judge choice and model response quality.

Pairwise agreement varies across judges, rang-

Judge pair	Spearman's ρ	Kendall's τ
Q235-Q30	0.60	0.40
Q235-GFL	0.80	0.60
Q235-G5N	1.00	1.00
Q30-GFL	0.40	0.40
Q30-G5N	0.60	0.40
GFL-G5N	0.80	0.60

Table 6: **Agreement between model rankings produced by different CRQ judges.** Q235: Qwen3-235B-Instruct; Q30: Qwen3-30B-Instruct; GFL: Gemini-2.5-Flash-Lite; G5N: GPT-5-Nano.

ing from 0.40 to 1.00 for both Spearman's ρ and Kendall's τ (Table 6). Although the judges identify the same top-performing model, the rankings of other models vary with judge choice. This result shows that CRQ evaluation can affect relative model comparisons as well as absolute scores.

Together, both converted benchmarks and *MyCulture* support the same conclusion. The measured competence is not only a property of the model, but it also depends on the evaluation instrument used to measure it.

3.3 Practical Instrument Selection

MCQs are suitable for broad, low-cost screening, initial model selection, and leaderboard comparison because they are easy to administer and score reproducibly. CRQs are appropriate when the evaluation target requires expressive, unconstrained responses and reliable human adjudication or multi-judge evaluation is available. CSRQs provide an auditable intermediate instrument: they remove exposed final-answer candidates while retaining deterministic scoring. We therefore recommend CSRQs as a matched audit alongside MCQ leaderboards when answer-option scaffolding may affect the interpretation of absolute scores. In low-resource settings, where reliable adjudication is costly or unavailable, their judge-free scoring can be particularly practical. When CRQ results are used for relative model comparisons, the observed variation across judges further motivates reporting multi-judge or judge-sensitivity analyses.

4 Benchmark Conversion Workflow

Converted benchmarks are used as supporting diagnostics for studying instrument sensitivity beyond *MyCulture*. To make these diagnostics reproducible, we document the conversion workflow used in our experiments. It transforms suit-

Model	OFE (%)	ACE (%)
GPT-5	2.5	97.5
Gemini-2.5-Pro	4.2	95.8
GLM-4.6	8.3	91.7
Qwen3-235B-Thinking	30.0	70.0
Average	11.25	88.75

Table 7: **Error analysis conducted on a subsample of incorrectly answered questions from each model.** OFE denotes output-format error, and ACE denotes answer content error.

able MCQ- and CRQ-based benchmark items into CSRQs under controlled conditions, making explicit how converted items were produced, filtered, and verified. We treat this workflow as reproducibility support for the empirical analysis, rather than as a claim that every benchmark item can be converted. Figure 6 summarizes the workflow.

Knowledge acquisition. Existing benchmark items are often written as single-answer MCQs or CRQs, and many cannot be directly reformulated as CSRQs. We therefore first extract the knowledge point required to answer the original item. An LLM with web-search augmentation identifies the relevant factual, relational, or procedural knowledge. This intermediate representation is used to preserve the intended knowledge while enabling structured CSRQ construction.

CSRQ-type selection and generation. Given the extracted knowledge point, an LLM-based classifier selects the most suitable CSRQ variant among multi-select, ordering, and matching. The classifier provides a short rationale for the selected variant. For ordering and matching items, an LLM generates the question prompt and the associated structured statements. For multi-select items, the LLM generates the question and candidate statements grounded in the extracted knowledge.

Statement verification. Generated statements are verified to reduce construction errors. For multi-select CSRQs, this includes checking both valid and distractor statements. Each statement is independently checked using an LLM with web-search capabilities. Statements that fail verification are discarded.

Reproducibility. We will release the prompts, converted items, filtering logs, and conversion code to enable reproducible analysis of instrument sensitivity in CSRQ evaluation.

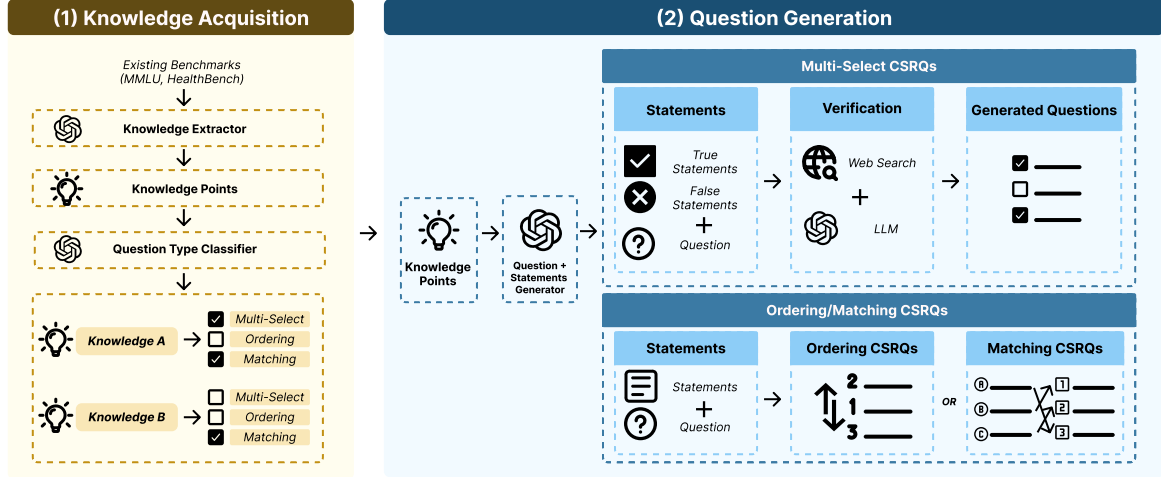


Figure 6: **CSRQ conversion workflow.** It converts suitable MCQ or CRQ items into CSRQ through knowledge acquisition, CSRQ-type selection, structured item generation, and verification. The workflow supports controlled analysis of instrument sensitivity under option-free elicitation and deterministic scoring.

Benchmark	Model	Orig.	G25P	G54
MMLU	GLM-4.6	84.80	83.63	87.72
	GPT-5	90.06	87.72	93.57
HBH	GLM-4.6	75.00	80.00	84.00
	GPT-5	77.00	88.00	86.00

Table 8: **Robustness of CSRQ evaluation to the LLM used for item construction.** Results are reported on matched subsets of MMLU and HealthBench-Hard (HBH). Orig. denotes the original GLM-4.6 construction pipeline; G25P denotes Gemini-2.5-Pro; and G54 denotes GPT-5.4.

4.1 Robustness to the CSRQ construction model.

Table 8 examines whether CSRQ results depend on the LLM used for item construction. We repeat the conversion procedure on matched subsets of MMLU and HealthBench-Hard using Gemini-2.5-Pro and GPT-5.4, alongside the original GLM-4.6 based pipeline. Although absolute scores vary across construction models, the relative ordering of the evaluated models remains unchanged: GPT-5 outperforms GLM-4.6 on both benchmarks under every construction pipeline. These results suggest that the comparative conclusions from the converted benchmarks are robust to the choice of LLM used for CSRQ construction.

5 Related Work

This work relates to four research lines, which are (a) MCQ-based LLM evaluation, (b) option-free

and answer-matching evaluation, (c) open-ended and (d) judge-based evaluation, and cultural LLM evaluation. We position CSRQ not as a new item archetype, but as an instrument-validity protocol that recomposes two useful functions, which are option-free elicitation and deterministic scoring.

MCQ-based LLM evaluation. MCQs are widely used in LLM benchmarks because they are scalable, reproducible, and objectively scored. Representative benchmarks include MMLU and its variants (Hendrycks et al., 2020; Li et al., 2023; Poh et al., 2024), as well as commonsense benchmarks such as HellaSwag (Zellers et al., 2019), CommonsenseQA (Talmor et al., 2019), and OpenBookQA (Mihaylov et al., 2018). However, MCQs expose final answer options. Prior work has noted that guessing can affect the reliability of multiple-choice assessment (Burton, 2001). Recent LLM evaluation studies further show that this issue is broader than random guessing alone. Answer options can create shortcuts, support elimination strategies, and change model behavior relative to option-free answering (Chandak et al., 2025; Raman et al., 2025; Balepur et al., 2025). In particular, reasoning models may benefit from seeing the option set before producing or finalizing their reasoning, leading to inflated MCQA performance compared with free-text response settings (Raman et al., 2025). We study this phenomenon as answer-option scaffolding, where the measured score partly reflects answer-option scaffolding rather than model competence alone.

Option-free and answer-matching evaluation.

A growing line of work (Chandak et al., 2025; Balepur et al., 2025; Hutchinson, 2026; Zhang and Nguyen, 2025) argues that LLM evaluation should move beyond conventional MCQs by asking models to generate answers without seeing final answer candidates. Answer-matching approaches replace option selection with free-form generation followed by reference-guided matching (Chandak et al., 2025). This direction is closely related to our motivation, since both settings aim to reduce the scaffolding introduced by visible answer options. However, CSRQ differs in its scoring target and response space. Rather than generating unconstrained natural-language answers and relying on answer matching, CSRQ constrains admissible responses into canonical structures such as subsets, orderings, and matchings. This allows option-free elicitation while preserving deterministic exact verification. Hence, CSRQ is not simply a free-form alternative to MCQs, but a middle-ground instrument between MCQ objectivity and CRQ option-free elicitation.

Option-free and answer-matching evaluation.

A growing line of work (Chandak et al., 2025; Balepur et al., 2025; Hutchinson, 2026; Zhang and Nguyen, 2025) evaluates LLMs without exposing final-answer candidates. Chandak et al. (Chandak et al., 2025) use free-form responses with a reference-guided LLM matcher, while BiGGen Bench (Kim et al., 2025) uses instance-specific rubrics and evaluator LMs for graded scoring. CSRQ differs by constraining responses to canonical structures, such as subsets, permutations, and matchings, enabling deterministic parsing and exact verification without an evaluation-time judge. Its contribution is therefore the combination of option-free elicitation and deterministic scoring, together with matched MCQ–CSRQ–CRQ comparisons of final-answer exposure and judge dependence

Structured constructed-response assessment.

Multi-select, ordering, and matching tasks have long existed in educational and psychological assessment (Haladyna, 2004; Nitko and Brookhart, 2006). CSRQ does not claim novelty in these item archetypes. Its contribution is to use such structured output spaces as canonical answer spaces for LLM evaluation. This allows the model to produce an answer without seeing final answer options, while correctness is determined by de-

terministic parsing and exact structural verification. Importantly, CSRQ removes the final answer-option layer, not all task-provided textual units. The statements or descriptions in a CSRQ define the task space, whereas MCQ options provide pre-composed candidate answers that may directly reveal the correct subset, order, or association. CSRQ therefore recomposes the useful scoring function of MCQs with the option-free elicitation function of CRQs.

Cultural LLM evaluation. Cultural benchmarks aim to evaluate models ability to capture culturally situated knowledge, including local norms, practices, languages, and values. Prior work has introduced culturally grounded benchmark datasets across diverse regional and linguistic contexts (Koto et al., 2024; Wang et al., 2024). Herein, we proposed *MyCulture* as the testbed.

6 Conclusion

This work frames LLM evaluation as an instrument-validity problem. MCQs offer deterministic scoring but may inflate measured competence through recognition, elimination, and partial-knowledge cues, while CRQs remove answer options but introduce grader-dependent variation. We introduce Constrained Structured Response Questions (CSRQs), which remove the final answer-option layer while preserving deterministic scoring. Across converted knowledge benchmarks and *MyCulture*, we show that measured performance changes with the evaluation format. The scores generally decrease when MCQ options are removed, while CRQ scores vary across judges. CSRQs therefore provide a middle ground between MCQ objectivity and CRQ option-free elicitation. Rather than replacing existing benchmarks, CSRQs offer an auditable instrument for studying how benchmark scores depend on evaluation protocols.

Limitation

CSRQs have several limitations. First, CSRQs are not universally applicable, as they require answers to be decomposable into at least two valid response units (e.g., selection, ordering, or matching), making some tasks difficult to construct or convert without reformulation. Second, CSRQs are more suitable for knowledge that can be represented through structured statements and may be less applicable to highly abstract, creative, or unconstrained reasoning tasks. Third, although CSRQs remove explicit answer options and reduce judge dependency, the response space remains finite and constrained by the provided statements, and therefore does not fully replicate open-ended generation as in CRQs. Finally, constructing CSRQs introduces additional benchmark development costs, including statement generation, distractor creation, and verification procedures to ensure correctness and consistency.

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Appendix

This appendix provides supplementary materials for the experiments and analyses presented in the main paper. We first present the chance-success analysis in Appendix A, followed by the evaluation prompts in Appendix B, representative *MyCulture* examples in Appendix C, and model artifacts in Appendix D. We then report the full results for benchmark conversion, *MyCulture* evaluation, CSRQ subtype sensitivity, and error analysis in Appendices E, F, G, and H, respectively.

A Chance-Success Analysis

Figure 7 visualizes chance-success baselines for MCQ, CSRQ variants, and an approximate CRQ reference baseline under exact structural scoring. The CRQ baseline is shown as near-zero because open-ended responses have an effectively unconstrained answer space under exact-match scoring, making random exact success negligible. This comparison is intended only as a structural baseline, such that it characterizes exact random success, not semantic difficulty, cognitive complexity, or genuine model understanding.

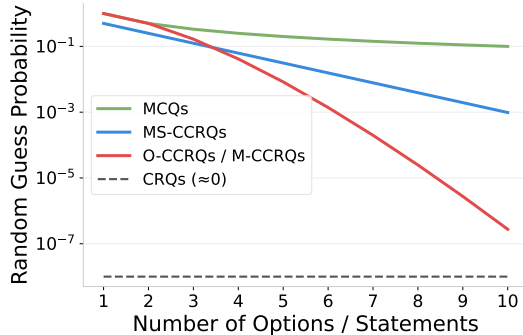


Figure 7: **Chance-success baselines under exact structural scoring.** The figure compares random-success probabilities for MCQs, CSRQ variants, and an approximate CRQ reference baseline as the number of options or statements increases. The CRQ baseline is shown as near-zero to reflect its open-ended answer space under exact-match scoring. The analysis characterizes chance success under exact scoring only, not semantic difficulty or cognitive complexity.

B Prompt Used in Experiments

Figure 8 and 9 illustrates the prompt used for MCQ evaluation in all experiment, Figure 17 present the prompts used for CRQ evaluation on *MyCulture* only. For HealthBench- Hard (Arora

et al., 2025) experiment, we follow their original prompt and judging method.

C Example of *MyCulture* dataset

In this section, we present representative examples from the *MyCulture* dataset, encompassing MCQs, CSRQs, and CRQs across three question structures, which are multi-select, ordering, and matching. Figures 11, 12, 13 illustrate example MCQs for the three structures, namely multi-select, ordering, and matching. Figures 14, 15, 16 present corresponding examples of CSRQs across the same three structures. Figure 17 provides an example of a CRQ.

D Model Artifacts

Table 11 summarizes the model version artifacts used during the experiment.

E Full performance for conversion of existing benchmark

This section shows full performance of model for experiment in Section 3 and Section 3.1.2. Table 12 show the performance of model on converting the existing multi-answer mcqs benchmark to CSRQ, while Table 13 show the performance of models benchmark of CSRQ generated from proposed conversion pipeline.

F Full performance for *MyCulture*

This section shows full performance of model for experiment in Section 3.2. Table 14 show the performance of model evaluated on *MyCulture* that consists 3 type of formats.

Scoring Strategies for CSRQs and CRQs.

Since both CSRQs and CRQs are option-free, we evaluate model responses using two scoring strategies, which are partial correctness and full correctness. The *loose* metric captures partial correctness, assigning credit when part of the predicted answer overlaps with the ground truth. The *strict* metric captures full correctness, requiring the predicted answer to exactly match the complete ground-truth sequence.

G CSRQ subtype sensitivity

Table 15 reports the performance degradation across the three question types Multi-Answer, Ordering, and Matching when transitioning from

Model	Category	Count	%	MS	O	M
GPT-5	Response Incompleteness	3	2.5	–	–	3
	Total	3	2.5	–	–	3
Gemini-2.5-Pro	Response Incompleteness	5	4.2	–	3	2
	Total	5	4.2	–	3	2
GLM-4.6	Response Incompleteness	10	8.3	–	–	10
	Total	10	8.3	–	–	10
Qwen3-235B-Thinking	Reasoning Truncation	30	25.0	6	13	11
	Response Incompleteness	5	4.2	–	–	5
	Instruction Following	1	0.8	–	–	1
	Total	36	30.0	6	13	17

Table 9: **Breakdown of output-format errors by remark category.** MS, O, and M denote Multi-Select, Ordering, and Matching question formats, respectively.

MCQ to CSRQ evaluation on *MyCulture*. An response incompleteness occurs when models do not generate all required matches, resulting in fewer matching pairs than the total number of statement pairs. In contrast, an instruction following error occurs when models fail to produce outputs in required formats. Reasoning truncation refers to cases where the response is truncated because the reasoning process exceeds the available output length.

H Error analysis on *MyCulture*

Table 9 breaks down output format errors by model and question format. Most parse errors come from Qwen3-235B-Thinking, mainly due to overly long reasoning outputs, while the other models mostly produce incomplete answers. The errors are concentrated in matching and ordering questions, with fewer issues observed in multi-select questions.

I Construction details of *MyCulture*

We describe the construction pipeline of *MyCulture* in Algorithm 1. We first scrape and curate textual materials from reliable sources such as Malaysia’s National Department for Culture and Arts (JKKK)¹ and relevant academic publications. We then use LLMs to draft question-answer pairs, across MCQ, CSRQ and CRQ formats. Additionally, we leverage LLMs to verify the format compliance with the target question type and evidence consistency, namely that the proposed answer is supported by the retrieved source content. For CSRQs, we additionally require the answer to be expressible in a canonical form that is deterministically parseable for objective scoring.

¹<https://pemetaanbudaya.jkkn.gov.my/>

Algorithm 1 Construction Pipeline for *MyCulture*

Require: Reliable cultural sources $\mathcal{S}$ (e.g., JKKK, academic journals), initial system prompt P_{sys}
Ensure: High-quality validated cultural dataset $\mathcal{D}_{\text{final}}$

- 1: **Step 1: Data Collection**
- 2: $C \leftarrow \text{ScrapeContent}(\mathcal{S})$
- 3: **while true do**
- 4: **Step 2: Question-Answer Generation**
- 5: $\mathcal{D}_{\text{raw}} \leftarrow \text{GenerateQA}(C, P_{\text{sys}})$
- 6: **Step 3: Automated Quality Filtering**
- 7: $\mathcal{D}_{\text{filtered}} \leftarrow \emptyset$
- 8: **for all** $(q, a) \in \mathcal{D}_{\text{raw}}$ **do**
- 9: $v_{\text{struct}} \leftarrow \text{JudgeStructure}(q)$
- 10: $v_{\text{fact}} \leftarrow \text{CheckAnswerability}(q, a)$
- 11: **if** $v_{\text{struct}} = \text{Valid}$ **and** $v_{\text{fact}} = \text{Correct}$ **then**
- 12: $\mathcal{D}_{\text{filtered}} \leftarrow \mathcal{D}_{\text{filtered}} \cup \{(q, a)\}$
- 13: **end if**
- 14: **end for**
- 15: **Step 4: Human Validation**
- 16: $\mathcal{D}_{\text{sample}} \leftarrow \text{Subsample}(\mathcal{D}_{\text{filtered}}, 5\%)$
- 17: $V_{\text{human}} \leftarrow \text{HumanValidate}(\mathcal{D}_{\text{sample}})$
- 18: **if** $V_{\text{human}} = \text{Pass}$ **then**
- 19: $\mathcal{D}_{\text{final}} \leftarrow \mathcal{D}_{\text{filtered}}$
- 20: **return** $\mathcal{D}_{\text{final}}$
- 21: **else**
- 22: $P_{\text{sys}} \leftarrow \text{RewritePrompt}(P_{\text{sys}})$
- 23: **end if**
- 24: **end while**

As an auxiliary sanity check, we run a small set of strong LLMs (e.g., Gemini-2.5-Pro, GPT-5, GLM-4.6, Qwen3-235B) on the filtered items to surface potentially ambiguous or ill-posed questions. We do not treat model failure as evidence of invalidity. Instead, flagged items are routed to manual inspection or discarded if they lack clear source support.

Finally, we conduct human auditing on a 5% subsample of the filtered dataset. Auditors are Malaysian residents with at least 10 years of lived experience in Malaysia, hold at least a bachelor degree, and have completed the national Malay examination (SPM)². We hire five auditors who assess whether each audited item is logical, culturally relevant, and answerable based on the provided source evidence. If the audit reveals recurring failure modes such as unsupported answers, ambiguous phrasing, or format violations, we refine the generation prompt and filtering rules and repeat the process. The hiring and audit process are reviewed by the Ethics Review Board (ERB), and all auditors are hired in compliance with Malaysia’s legal minimum wage requirements. **In total we have constructed 5,463 question, where each type of question MCQ, CSRQ and CRQ get 1821 question.**

²https://ms.wikipedia.org/wiki/Sijil_Pelajaran_Malaysia

J Performance of Models on various Judge

Table 10 show the detailed performance across different judges. The experiment is discuss at Section 3.1.2.

Evaluated Model	Judge Model			σ
	Qwen-235B-Instruct	GPT-5.2	Gemini-2.5-Pro	
GPT-5	50.84	34.60	54.85	10.72
Gemini-2.5-Pro	31.07	9.80	33.12	12.91
GLM-4.6	32.34	9.66	34.79	13.86
Qwen3-235B-Thinking	41.18	14.63	43.94	16.18
Average	–	–	–	13.42

Table 10: **CRQ evaluation scores under different LLM judges.** Higher standard deviation (σ) indicates greater sensitivity to judge selection. All models are evaluated with reasoning mode.

Malay Prompt

Berikut adalah soalan. Format jawapan mengikut jenis soalan adalah seperti berikut: bagi soalan jenis Gabungan, gunakan format jawapan \boxed{A}, \boxed{B}, \boxed{C}, atau \boxed{D}; bagi soalan jenis Berurutan, gunakan format jawapan \boxed{A}, \boxed{B}, \boxed{C}, atau \boxed{D}; dan bagi soalan jenis Padanan, gunakan format jawapan \boxed{A}, \boxed{B}, \boxed{C}, atau \boxed{D}. Berikut adalah soalan: {soalan}. Jenis soalan: {jenissoalan}. Jawapan:

English Prompt

Below is a question. The answer format according to the question type is as follows: for Combination questions, use the answer format \boxed{A}, \boxed{B}, \boxed{C}, or \boxed{D}; for Ordering questions, use the answer format \boxed{A}, \boxed{B}, \boxed{C}, or \boxed{D}; and for Matching questions, use the answer format \boxed{A}, \boxed{B}, \boxed{C}, or \boxed{D}. The question is as follows: {question}. Question type: {questiontype}. Answer:

Figure 8: **Prompt used for the all MCQ experiment.** The upper prompt (in Malay) shows the Malay prompt, while the lower prompt presents its English version.

Malay Prompt

Berikut adalah soalan. Jawapan perlu diletakkan dalam format \boxed{}. Format jawapan mengikut jenis soalan adalah seperti berikut: bagi soalan jenis Gabungan, gunakan format \boxed{A,B,C,D}; bagi soalan jenis Berurutan, gunakan format \boxed{A,B,C,D}; dan bagi soalan jenis Padanan, gunakan format \boxed{[[A,i],[B,ii],[C,iii],...]}. Berikut adalah soalan: {soalan}. Jenis soalan: {jenissoalan}. Jawapan:

English Prompt

Below is a question. The answer must be provided in the \boxed{} format. According to the question type, the answer format is as follows: For Combination questions, use the format \boxed{A,B,C,D}; For Ordering questions, use the format \boxed{A,B,C,D}; For Matching questions, use the format \boxed{[[A,i],[B,ii],[C,iii],...]}.

The question is as follows: {question}. Question type: {questiontype}. Answer:

Figure 9: **Prompt used for the all CSRQs experiment.** The upper prompt (in Malay) shows the Malay prompt, while the lower prompt presents its English version.

Prompt
Sila jawab soalan berikut: {soalan} Jawapan:

Translated Prompt
Please answer the following question: {question} Answer:

Figure 10: **Prompt used for the CRQs experiments *MyCulture*.** The upper prompt (in Malay) shows the original prompt, while the lower prompt presents its translated version.

Multi Select - MCQs

Original Question
Apakah kenyataan yang betul mengenai permainan Lansaran dan budaya suku kaum Murut?
Pernyataan:
I. Lansaran dibina secara individu tanpa melibatkan masyarakat.
II. Permainan Lansaran memerlukan ukuran kawasan antara 8 hingga 30 kaki persegi.
III. Kayu 'Selangan Batu' digunakan dalam pembinaan lantai Lansaran kerana ketahanannya.
IV. Lansaran hanya untuk kanak-kanak dan tidak melibatkan orang dewasa.
Pilihan:
A. II, III
B. I, IV
C. I, II, IV
D. II, IV
Jawapan: A

Translated Question
What statements are correct regarding the Lansaran game and the culture of the Murut community?
Statements:
I. The Lansaran is built individually without involving the community.
II. The Lansaran game requires an area size of between 8 and 30 square feet.
III. "Selangan Batu" wood is used in constructing the Lansaran floor due to its durability.
IV. The Lansaran is only for children and does not involve adults.
Options:
A. II, III
B. I, IV
C. I, II, IV
D. II, IV
Answer: A

Figure 11: **Example of a multi-select MCQ.** The left panel presents the original question, while the right panel shows its translated counterpart.

Ordering - MCQs

Original Question
Pilih urutan yang betul bagi proses berikut berkaitan dengan penghasilan kuih tradisional:
Pernyataan:
I. Uli tepung gandum dan basuh tepung untuk membuang gluten.
II. Proses membasuh tepung bertujuan menghasilkan kanji.
III. Air tepung diperap dan buang air setiap 6 jam selama 3 hari.
IV. Masukkan tepung yang mendap ke dalam air gula dan kacau.
Pilihan:
A. I, II, III, IV
B. II, I, IV, III
C. III, I, IV, II
D. IV, III, II, I
Jawapan: A

Translated Question
Select the correct order for the following process related to the production of traditional delicacy:
Statements:
I. Knead the wheat flour and wash it to remove gluten.
II. The process of washing the flour aims to produce starch.
III. The flour water is left to ferment, and the water is discarded every 6 hours for 3 days.
IV. Add the settled flour into sugar water and stir.
Options:
A. I, II, III, IV
B. II, I, IV, III
C. III, I, IV, II
D. IV, III, II, I
Answer: A

Figure 12: **Example of a ordering MCQ.** The left panel presents the original question, while the right panel shows its translated counterpart.

Matching - MCQs

Original Question	Translated Question
Padankan alat muzik dengan peranannya. Kumpulan Satu: A. Kertuk Ulu B. Tamborin C. Rebana D. Canang Kumpulan Dua: I. Menambah ritma dan variasi bunyi. II. Memberikan bunyi gabungan gitar bass. III. Sebagai penyeri irama di bahagian klimaks. IV. Sebagai lapisan ritma tambahan. Pilihan: A. A-II, B-I, C-IV, D-III B. A-I, B-IV, C-II, D-III C. A-IV, B-I, C-III, D-II D. A-II, B-III, C-IV, D-I Jawapan: C	Match the musical instruments with their roles. Group One: A. Kertuk Ulu B. Tambourine C. Rebana D. Canang Group Two: I. Adds rhythm and sound variation. II. Produces a sound combination similar to a bass guitar. III. Serves as a rhythmic embellishment during the climax section. IV. Acts as an additional rhythmic layer. Options: A. A-II, B-I, C-IV, D-III B. A-I, B-IV, C-II, D-III C. A-IV, B-I, C-III, D-II D. A-II, B-III, C-IV, D-I Answer: C

Figure 13: **Example of a matching MCQ.** The left panel presents the original question, while the right panel shows its translated counterpart.

Multi Select - CCRQs

Original Question	Translated Question
Apakah kenyataan yang betul mengenai permainan Lansaran dan budaya suku kaum Murut? Pernyataan: A. Lansaran dibina secara individu tanpa melibatkan masyarakat. B. Permainan Lansaran memerlukan ukuran kawasan antara 8 hingga 30 kaki persegi. C. Kayu 'Selangan Batu' digunakan dalam pembinaan lantai Lansaran kerana ketahanannya. D. Lansaran hanya untuk kanak-kanak dan tidak melibatkan orang dewasa. Jawapan: B,C	What statements are correct regarding the Lansaran game and the culture of the Murut community? Statements: A. The Lansaran is built individually without involving the community. B. The Lansaran game requires an area size of between 8 and 30 square feet. C. "Selangan Batu" wood is used in constructing the Lansaran floor due to its durability. D. The Lansaran is only for children and does not involve adults. Answer: B,C

Figure 14: **Example of a multi-select CSRQ.** The left panel presents the original question, while the right panel shows its translated counterpart.

Ordering - CCRQs

Original Question	Translated Question
Pilih urutan yang betul bagi proses berikut berkaitan dengan penghasilan kuih tradisional: Pernyataan: A. Uli tepung gandum dan basuh tepung untuk membuang gluten. B. Proses membasuh tepung bertujuan menghasilkan kanji. C. Air tepung diperap dan buang air setiap 6 jam selama 3 hari. D. Masukkan tepung yang mendap ke dalam air gula dan kacau. Jawapan: A, B, C, D	Select the correct order for the following process related to the production of traditional delicacy: Description: A. Knead the wheat flour and wash it to remove gluten. B. The process of washing the flour aims to produce starch. C. The flour water is left to ferment, and the water is discarded every 6 hours for 3 days. D. Add the settled flour into sugar water and stir. Answer: A, B, C, D

Figure 15: **Example of a ordering CSRQ.** The left panel presents the original question, while the right panel shows its translated counterpart.

Matching - CCRQs

Original Question	Translated Question
Padankan alat muzik dengan peranannya. Kumpulan Satu: A. Kertuk Ulu B. Tamborin C. Rebana D. Canang Kumpulan Dua: I. Menambah ritma dan variasi bunyi. II. Memberikan bunyi gabungan gitar bass. III. Sebagai penyeri irama di bahagian klimaks. IV. Sebagai lapisan ritma tambahan. Jawapan: A-IV, B-I, C-III, D-II	Match the musical instruments with their roles. Group One: A. Kertuk Ulu B. Tambourine C. Rebana D. Canang Group Two: I. Adds rhythm and sound variation. II. Produces a sound combination similar to a bass guitar. III. Serves as a rhythmic embellishment during the climax section. IV. Acts as an additional rhythmic layer. Answer: A-IV, B-I, C-III, D-II

Figure 16: **Example of a matching CSRQ.** The left panel presents the original question, while the right panel shows its translated counterpart.

CRQs

Original Question	Translated Question
Huraikan ciri-ciri fizikal, cara pembinaan dan peranan komuniti dalam permainan tradisional Lansaran suku kaum Murut berdasarkan amalan tradisi yang dinyatakan. Titik jawapan: • Tempat permainan Lansaran berukuran antara 8 hingga 30 kaki persegi. • Lansaran dapat menampung lebih kurang 30 orang pemain. • Lansaran dibina secara bergotong-royong di ruang khas rumah panjang suku kaum Murut. • Kayu 'Selangan Batu' digunakan kerana keras dan liat. • Kayu diikat dengan rotan saga untuk menampung lantai. • Lantai dibina sekaki ke bawah secara tergantung.	Describe the physical characteristics, construction methods, and the role of the community in the traditional Lansaran game of the Murut ethnic group based on the stated traditional practices. Answer points: • The Lansaran playing area measures between 8 and 30 square feet. • The Lansaran can accommodate approximately 30 players. • The Lansaran is constructed through communal cooperation (gotong-royong) in a designated space within the Murut longhouse. • Selangan Batu wood is used because it is hard and resilient. • The wood is bound with rotan saga to support the floor structure. • The floor is built suspended, approximately one foot above the ground.

Figure 17: **Example of a CRQ.** The left panel presents the original question, while the right panel shows its translated counterpart.

Model	Artifact / Version
GPT-5	gpt-5-2025-08-07
GPT-5-mini	gpt-5-mini-2025-08-07
GPT-5-nano	gpt-5-nano-2025-08-07
Gemini-2.5-Pro*	latest
Gemini-2.5-Flash*	latest
Gemini-2.5-Flash-Lite*	latest
GLM-4.6*	latest
GLM-4-32B-128K	0414
Qwen3-235B-Thinking	Qwen/Qwen3-235B-A22B-Thinking-2507
Qwen3-30B-Thinking	Qwen/Qwen3-30B-A3B-Thinking-2507
Qwen3-4B-Thinking	Qwen/Qwen3-4B-Thinking-2507
Qwen3-235B-Instruct	Qwen/Qwen3-235B-A22B-Instruct-2507
Qwen3-30B-Instruct	Qwen/Qwen3-30B-A3B-Instruct-2507
Qwen3-4B-Instruct	Qwen/Qwen3-4B-Instruct-2507
SeaLLM-v3-7B	SeaLLMs/SeaLLMs-v3-7B-Chat
SeaLLM-v3-1.5B	SeaLLMs/SeaLLMs-v3-1.5B-Chat
Qwen-SEA-LION-V4-32B-IT	aisingapore/Qwen-SEA-LION-v4-32B-IT
MaLLaM-1.1B	mesolitica/mallam-1.1b-20k-instructions-v2
MaLLaM-5B	mesolitica/mallam-5b-20k-instructions-v2

Table 11: **Model variants and their associated artifact versions used in the experiments are reported.** For models without explicit release identifiers, the most recent publicly available version at the time of evaluation was adopted. An asterisk (*) denotes models that do not disclose artifact details; in such cases, the version corresponds to the latest available release as of May 2026.

Family	Model	Reasoning	MalayMMLU			CS-EN			Avg. Δ
			MCQ	CSRQ	Δ	MCQ	CSRQ	Δ	
Closed-source Models									
OpenAI	GPT-5 [†]	✓	81.98	59.46	22.52	76.32	63.16	13.16	17.84
	GPT-5-mini [†]	✓	78.08	50.45	27.63	69.74	47.37	22.37	25.00
	GPT-5-nano [†]	✓	67.87	38.44	29.43	64.47	40.79	23.68	26.56
Google	Gemini-2.5-Pro [†]	✓	84.38	56.16	28.22	80.26	51.32	28.94	28.58
	Gemini-2.5-Flash [†]	✓	63.06	48.05	15.01	69.74	51.32	18.42	16.72
	Gemini-2.5-Flash-Lite [†]	✓	64.86	37.84	27.02	60.53	43.42	17.11	22.07
Open-source Models									
Z.ai	GLM-4.6 [†]	✓	73.87	47.15	26.72	75.00	46.05	28.95	27.84
			67.57	38.44	29.13	68.42	44.74	23.68	26.41
Alibaba	Qwen3-235B-Thinking [†]	✓	77.18	45.65	31.53	76.32	57.89	18.43	24.98
	Qwen3-30B-Thinking [†]	✓	71.77	34.23	<u>37.54</u>	73.68	44.74	<u>28.94</u>	<u>33.24</u>
	Qwen3-4B-Thinking [†]	✓	63.96	25.53	38.43	65.79	46.05	19.74	29.09
	Qwen3-235B-Instruct		76.88	45.05	31.83	86.84	63.13	23.71	27.77
	Qwen3-30B-Instruct		70.57	35.14	35.43	82.89	44.74	38.15	36.79
	Qwen3-4B-Instruct		58.86	24.32	34.54	65.75	39.47	26.28	30.41
Average Δ			29.64			23.68			26.66

Table 12: **Option-removal effect on existing benchmarks.** For each benchmark, CSRQ removes final answer options while preserving objective exact-match scoring. We report $\Delta = \text{MCQ} - \text{CSRQ}$ to make the effect of removing answer candidates explicit. Across MalayMMLU and CS-EN, models consistently score lower under CSRQ, with an average drop of 26.66 points. Bold indicates the largest degradation and underlined values indicate the second-largest degradation. [†] denotes reasoning models and by default we set the reasoning effort to medium where available.

Family	Model	Reasoning	MMLU			HealthBench-Hard		
			MCQ	CSRQ	Δ	CRQ	CSRQ	Δ
Closed-source Models								
OpenAI	GPT-5 [†]	✓	92.50	84.64	7.86	50.84	83.33	32.49
	GPT-5-mini [†]	✓	<u>90.51</u>	<u>82.30</u>	8.21	<u>48.89</u>	<u>82.51</u>	33.62
	GPT-5-nano [†]	✓	87.01	72.02	14.99	<u>42.44</u>	72.95	30.51
Google	Gemini-2.5-Pro [†]	✓	<u>91.06</u>	78.35	12.71	31.07	75.68	<u>44.61</u>
	Gemini-2.5-Flash [†]	✓	87.61	80.02	7.59	29.82	<u>78.42</u>	48.60
	Gemini-2.5-Flash-Lite [†]	✓	86.80	74.49	12.31	22.96	59.84	36.88
Open-source Models								
Z.ai	GLM-4.6 [†]	✓	84.38	80.59	3.79	32.34	77.87	<u>45.53</u>
			81.30	72.16	9.14	32.21	69.95	37.74
Alibaba	Qwen3-235B-Thinking [†]	✓	90.39	80.90	8.61	41.18	<u>78.42</u>	37.24
	Qwen3-4B-Thinking [†]	✓	81.58	64.15	17.43	23.06	58.20	35.14
	Qwen3-30B-Thinking [†]	✓	87.72	73.05	14.67	31.59	71.31	39.72
	Qwen3-235B-Instruct		89.51	81.66	7.85	31.87	77.60	45.73
	Qwen3-30B-Instruct		85.55	73.91	11.64	27.95	71.86	43.91
	Qwen3-4B-Instruct		79.41	62.32	<u>17.09</u>	22.52	59.02	36.50
Average Δ					10.88	39.25		

Table 13: **Performance comparison between original multiple-choice evaluation and CSRQ evaluation across MMLU and HealthBench-Hard.** Δ denotes the absolute performance degradation between evaluation settings. Bold indicates the largest degradation and underlined values indicate the second-largest degradation within each Δ column. Qwen3-235B-A22B-Instruct is used as judge for CRQ. [†] denotes reasoning models and by default we set the reasoning effort to medium where available.

Family	Model	Reasoning	MCQs	CSRQs		CRQs	
				loose	strict	loose	strict
	Random Baseline		25.53	40.33	4.28	-	-
OpenAI	GPT-5 [†]	✓	88.14	78.10	56.78	68.41	22.90
	GPT-5-mini [†]	✓	84.73	77.50	55.46	67.25	25.65
	GPT-5-nano [†]	✓	80.12	73.61	48.38	63.10	20.76
	GPT-4o		90.28	<u>78.00</u>	<u>56.01</u>	68.61	21.14
	GPT-4o-mini		78.36	74.30	49.20	65.78	20.37
Google	Gemini-2.5-Pro [†]	✓	85.56	77.78	55.68	66.87	20.26
	Gemini-2.5-Flash [†]	✓	82.87	71.05	47.39	65.23	21.47
	Gemini-2.5-Flash-Lite [†]	✓	79.57	72.90	48.00	62.56	18.95
Z.ai	GLM-4.6 [†]	✓	84.46	76.37	53.38	63.20	17.74
	GLM-4.6 [†]		87.37	77.45	54.91	56.72	17.68
	GLM-4-32B-128K		75.18	71.88	47.83	59.26	16.09
Alibaba	Qwen3-235B-Thinking [†]	✓	61.01	62.62	42.12	50.13	18.51
	Qwen3-30B-Thinking [†]	✓	79.57	70.49	43.82	62.48	17.35
	Qwen3-4B-Thinking [†]	✓	71.61	66.42	37.67	48.47	10.98
	Qwen3-235B-Instruct		80.12	73.55	51.29	62.93	18.34
	Qwen3-30B-Instruct		82.48	75.21	52.50	59.03	17.13
	Qwen3-4B-Instruct		76.06	65.64	35.48	55.97	15.27
SeaLLM	SeaLLMs-v3-7B		29.98	52.96	26.30	47.62	7.86
	SeaLLMs-v3-1.5B*		0.00	0.16	0.00	39.54	6.45
AI SG	Qwen-SEA-LION-v4-32B-IT		83.25	76.16	53.54	62.55	18.23
Mesolitica	MaLLaM-1.1B*		0.00	0.00	0.00	39.56	5.44
	MaLLaM-5B*		0.00	0.00	0.00	46.10	6.64

Table 14: **Performance of various LLMs on the *MyCulture* benchmark across MCQ, CSRQ, and CRQ tasks.** **Bold** values indicate the best performance, while underlined values denote the second-best. An asterisk (*) marks instances where the model failed to follow instructions to output the answer in a parseable format, hence unable to be extracted via regular expressions. † denotes reasoning models and by default we set the reasoning effort to medium where available.

Family	Model	Multi Select			Ordering			Matching			Overall		
		MCQ	CSRQ	Δ	MCQ	CSRQ	Δ	MCQ	CSRQ	Δ	MCQ	CSRQ	Δ
OpenAI	GPT-5 [†]	89.00	68.57	20.43	89.04	50.51	<u>38.53</u>	86.03	48.23	37.80	88.14	56.78	31.36
	GPT-5-mini [†]	89.71	66.71	23.00	82.53	48.63	33.90	80.63	48.23	32.40	84.73	55.46	29.27
	GPT-5-nano [†]	85.57	50.57	35.00	79.97	47.43	32.54	73.18	46.55	26.63	80.12	48.38	31.74
	GPT-4o	92.14	71.00	21.14	91.27	47.95	43.32	86.78	45.25	41.53	90.28	56.01	34.27
	GPT-4o-mini	84.71	59.14	25.57	77.57	46.40	31.17	70.95	39.29	31.66	78.36	49.20	29.16
Google	Gemini-2.5-Pro [†]	90.00	67.71	22.29	82.53	48.63	33.90	83.05	47.67	35.38	85.56	55.68	29.88
	Gemini-2.5-Flash [†]	86.71	52.43	34.28	78.94	46.40	32.54	82.12	41.90	<u>40.22</u>	82.87	47.39	<u>35.48</u>
	Gemini-2.5-Flash-Lite [†]	82.71	52.00	30.71	76.88	47.09	29.79	78.40	43.76	34.64	79.57	48.00	31.57
Z.ai	GLM-4.6 [†]	90.43	65.57	24.86	79.11	48.63	30.48	82.50	42.64	39.86	84.46	53.38	31.08
	GLM-4.6 [†]	94.14	65.43	28.71	88.53	52.40	36.13	77.28	43.95	33.33	87.37	54.91	32.46
	GLM-4-32B-128K	87.71	63.14	24.57	75.17	44.52	30.65	58.85	31.47	27.38	75.18	47.83	27.35
Alibaba	Qwen3-235B-Thinking [†]	72.29	45.71	26.58	60.62	42.98	17.64	46.74	36.50	10.24	61.01	42.12	18.89
	Qwen3-30B-Thinking [†]	82.14	42.00	40.14	80.82	48.63	32.19	74.86	40.97	33.89	79.57	43.82	<u>35.75</u>
	Qwen3-4B-Thinking [†]	74.29	31.43	<u>42.86</u>	76.20	46.40	29.80	63.13	36.31	26.82	71.61	37.67	33.94
	Qwen3-235B-Instruct	88.43	62.57	25.86	81.16	46.75	34.41	68.16	41.53	26.63	80.12	51.29	28.83
	Qwen3-30B-Instruct	88.71	63.86	24.85	80.99	48.63	32.36	75.98	41.90	34.08	82.48	52.50	29.98
	Qwen3-4B-Instruct	81.43	38.00	43.43	77.57	39.55	38.02	67.41	27.75	39.66	76.06	35.48	40.58
SeaLLM	SeaLLMs-v3-7B	33.43	33.86	-0.43	50.51	29.97	20.54	3.17	12.48	-9.31	29.98	26.30	3.68
	SeaLLMs-v3-1.5B	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
AI SG	Qwen-SEA-LION-v4-32B	90.00	66.43	23.57	80.14	49.32	30.82	77.84	41.34	36.50	83.25	53.54	29.71
Mesolitica	MaLLaM-5B	0	0	0	0	0	0	0	0	0	0	0	0
	MaLLaM-1.1B	0	0	0	0	0	0	0	0	0	0	0	0
Average Δ				26.41	32.16			<u>31.31</u>			30.39		

Table 15: **Performance comparison between MCQ and CSRQ settings across Multi-Select, Ordering, Matching, and overall evaluation on MyCulture.** Δ denotes absolute performance degradation from MCQ to CSRQ. Within each Δ column, **bold** indicates the largest degradation and underlined values indicate the second-largest. [†] denotes reasoning models and by default we set the reasoning effort to medium where available.